MODULE 28 · DOMAIN 4 - ASSEMBLING AND DEPLOYING APPLICATIONS

Interfaces: Apps, Slack and Teams

Putting the application in front of users, with the right identity, in the tools they already use.

SECTION 4 of 6 EXAM WEIGHT 22% MODULE 28 of 40

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Meet users where they already work. The best interface is the one people already have open.

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Ways to put an application in front of users

INTERFACE	BEST FOR
Chat UI from an agent template	A ready chat window on Databricks Apps, quickly
Custom Databricks App	Your own web UI: Streamlit, Gradio, Dash, React and others
Genie app in Slack or Teams	Data questions inside the chat tools teams already use
Endpoint or API	Your own front end or another system calling the agent
AI Playground and Review App	Internal testing and expert feedback, not end users

ON THE EXAM

The Playground and Review App are for builders and experts. End users need an app, a chat tool, or your own front end.

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App authorisation, or user authorisation?

APP AUTHORISATION

The app's own identity

- A dedicated service principal per app
- Every user shares its permissions
- Resources declared in databricks.yml
- Simple, the default

USER AUTHORISATION

The user's identity

- The user's token is forwarded to the app
- Unity Catalog policies apply per user
- Limited by the scopes the app requests
- Public Preview, enabled by an admin

THE TIEBREAKER

The same rule as module 25: if users must see different data, use the user's identity. Otherwise, the app's.

How it works in practice

- The user's token arrives in the **x-forwarded-access-token** request header.
- **Scopes**, such as sql, genie and model-serving, limit what the app may do.
- Anything outside the approved scopes is blocked, even if the user could do it elsewhere.
- Users consent to the scopes the first time they open the app.
- Supervisor Agent endpoints **always** require user authorisation.

> ON THE EXAM
>
> Request only the scopes the app needs. Least privilege applies to apps acting on a user's behalf.

Getting a chat interface running

- Start from an **agent template**. Its built-in chat UI streams, renders markdown and handles sign-in.
- The production chat template is **e2e-chatbot-app-next**. The older Streamlit one lacks its features.
- Optional **persistent chat history**, using the memory ideas from module 26.
- Grant users permission to use the app, then share its URL.
- Long-running agent tasks need care: apps have a request timeout, so use background mode.

> ON THE EXAM
>
> For an internal chat window on a custom agent, the template's chat UI is the least-effort answer.

Genie in Slack or Teams, or a custom app?

GENIE APP IN SLACK OR TEAMS

Data questions, no code

- Questions over Genie Agents
- Inside the team's chat tool
- Install and connect, no build
- User's own permissions via SSO

CUSTOM APP

Your own agent, your own UI

- A custom agent's behaviour
- Branded or specialised screens
- Workflows beyond data questions
- You build and maintain it

THE TIEBREAKER

If the need is data questions where the team already chats, install the Genie app. Build only when the agent or UI is your own.

What the Genie apps do

- Users send a **direct message**, or mention **@Genie** in a channel.
- A channel owner can **pin a specific Genie Agent** to the channel.
- Sign-in uses **SSO**, so each user's existing data permissions are enforced.
- The audit log records whether each request came from Slack, Teams or the web.
- The Slack app is in Public Preview, enabled by an account admin.

> ON THE EXAM
>
> Pinning a Genie Agent to a channel gives that team consistent answers from the right data domain.

Chatting in Slack, or posting to Slack?

AN INTERFACE IN SLACK

Users talk to the agent

- People ask questions in Slack
- The agent replies there
- The Genie app, or a custom bot
- Slack is the front door

A SLACK TOOL

The agent sends a message

- The agent decides to notify someone
- It calls Slack as a tool action
- Through a Unity Catalog connection
- Slack is a destination

THE TIEBREAKER

If the agent must alert a channel on its own initiative, Slack is a tool, reached through a Unity Catalog connection, not an interface.

Pennant Outdoor's interfaces

01

Customers

Custom web app: branded chat, returns flow.

02

Staff

Agent template chat UI, app authorisation.

03

Regional managers

App with user authorisation: own region only.

04

Sales team

Genie app in Teams, pinned sales Genie Agent.

05

Stock alerts

Agent posts to Slack through a UC connection.

ON THE EXAM

Five audiences, five front doors, one set of governed agents and data behind them.

> A team has built a ResponsesAgent and wants internal staff to chat with it in a browser, with streaming replies, as soon as possible.
>
> All staff may see the same data.

What is the least-effort way to deliver this?

A Build a React front end from scratch and host it elsewhere

B Deploy from an agent template on Apps, using its chat UI

C Ask all the staff to use the agent in the AI Playground

D Give staff the endpoint URL and a curl command to try it

◆ PAUSE AND DECIDE

What is the least-effort way to deliver this?

A React from scratch

Most effort, and hosting outside Databricks governance.

B Template chat UI on Apps

Built-in streaming chat, sign-in and sharing, ready now.

C AI Playground

A builder's testing tool, not an end-user interface.

D Endpoint and curl

Not a chat interface for staff.

ON THE EXAM

Browser chat, streaming, as soon as possible, same data for all. The template chat UI with app authorisation fits exactly.

> A Databricks App shows sales figures from a table that has Unity Catalog row filters by region.
>
> Each manager must see only their own region, and the app must not reimplement the filtering.

How should the app authenticate to the data?

A App authorisation, since the service principal is simpler

B App authorisation, filtering the rows in the app's Python code

C One personal access token shared by everyone using the app

D User authorisation with the sql scope and the user's token

◆ PAUSE AND DECIDE

How should the app authenticate to the data?

A App authorisation

Everyone sees what the service principal sees.

B App auth, filter in Python

Reimplements the filter, which the scenario forbids.

C Shared personal token

One identity for all, and a leaked long-lived secret.

D User authorisation, sql scope

The existing row filters apply to each manager.

ON THE EXAM

Existing row filters, per-user results, no reimplementing. The user's own identity is the only way the filter applies.

> A sales team spends its day in Microsoft Teams. It wants to ask questions about pipeline and bookings data, which a Genie Agent already covers.
>
> The data team wants the smallest possible build.

What should the data team do?

A. Install the Databricks Genie app in Microsoft Teams

B. Build a custom Teams bot that calls the Genie REST API

C. Email the Genie answers to the team every morning

D. Deploy a Knowledge Assistant over the sales tables

◆ PAUSE AND DECIDE

What should the data team do?

A Genie app in Teams
Data questions in Teams, no build, users' own permissions.

B Custom Teams bot
Builds what the Genie app already provides.

C Morning email
Not interactive. Cannot answer new questions.

D Knowledge Assistant
For documents. Genie already covers the tables.

ON THE EXAM

Data questions, a Genie Agent that exists, a team in Teams, smallest build. Install the Genie app and pin the agent.

> A claims agent scores every new claim. When a claim is high risk, the fraud team must be alerted in its Slack channel straight away.
>
> Nobody on the fraud team will be chatting with the agent.

How should the alert be delivered?

A Install the Databricks Genie app for Slack in the channel

B Ask the fraud team to watch the agent's chat UI all day

C Give the agent a Slack tool through a UC connection

D Email the fraud team a daily digest of high-risk claims

◆ PAUSE AND DECIDE

How should the alert be delivered?

A Genie app for Slack

For people asking questions. Nobody is asking.

B Watch the chat UI

Not an alert, and not realistic.

C Slack tool via UC connection

The agent posts on its own initiative, governed.

D Daily email digest

Fails straight away.

ON THE EXAM

The agent must notify, and nobody is chatting. Slack is a destination, so it is a tool, not an interface.

What to carry forward

- Meet users where they work: **template chat UI**, custom app, **Genie in Slack or Teams**, or an API.
- **App authorisation**: one service principal. **User authorisation**: user's token, limited by scopes.
- Supervisor Agent endpoints always need user authorisation.
- Data questions in chat tools: install the **Genie app**. Build only for your own agent or UI.
- Alerting a channel is a **tool**, not an interface. Playground is not for end users.

> ON THE EXAM
>
> That completes section 4. Next, section 5: module 29 defends against malicious input.